

Short Paper:

Earned Value Analysis and Percent Completion Methods

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Introduction

Earned Value Analysis is a core project management technique that integrates cost, schedule, and scope to evaluate project performance. This short paper explains the purpose and components of earned value analysis, outlines the three primary methods for estimating percent completion, and demonstrates these concepts through two exercises from the course textbook. The paper then applies earned value analysis to selected milestones from the redevelopment of the Sanford B. Ladd School, using realistic project data to illustrate how different percent completion methods influence performance measurement. Together, these sections show how EVA supports informed decision making and effective monitoring throughout the life cycle of a complex community redevelopment project.

Earned Value Analysis Overview

Earned Value Analysis is a performance measurement approach that compares the budgeted cost of scheduled work, the budgeted cost of completed work, and the actual cost of that work (Meredith et al., 2021). Planned Value represents the authorized budget for scheduled activities. Earned Value represents the budgeted cost of the work actually completed. Actual Cost represents the real expenditures incurred. From these values, project managers calculate cost variance, schedule variance, the Cost Performance Index, and the Schedule Performance Index to determine whether a project is ahead or behind schedule and over or under budget. EVA is especially valuable for complex, multi-phase projects like the redevelopment of the Sanford B. Ladd School because it provides early insight into performance trends and supports data driven decision making.

Methods for Estimating Percent Completion

Project managers use several methods to determine how much value has been earned at a given point in time. The three most common methods are the 0/100 rule, the 50/50 rule, and the percent complete method (Microsoft, n.d.). Percent complete values directly influence earned value calculations because they determine how much of the budgeted work is considered complete at any point in time, which affects both cost and schedule performance indicators.

The 0/100 rule assigns zero percent earned value until a task is fully complete. This method is appropriate for binary deliverables that cannot be partially credited. For the Ladd School project, obtaining the certificate of occupancy is a suitable example because the building is either approved for use or it is not. This method prevents premature recognition of progress and ensures that value is only recorded when the deliverable is fully achieved.

The 50/50 rule assigns fifty percent earned value at the start of a task and the remaining fifty percent at completion. This method is useful for documentation-based tasks that have clear start and end points. Preparing the HUD Community Development Block Grant application is an appropriate example because the work is divided into preparation and submission stages. The 50/50 method simplifies tracking while still recognizing meaningful progress.

The percent complete method assigns earned value based on actual measured progress. This method is appropriate for long duration construction activities where progress can be quantified. Masonry and façade repair at the Ladd School is a strong example because progress can be measured by square footage or sections completed (Drexel Tech, 2026; LocalCode, n.d.). This method provides the most accurate representation of work performed.

Textbook Problem Solutions

The following section applies earned value concepts to two exercises from the textbook (Meredith et al., 2021) and then demonstrates how each percent completion method can be used to evaluate progress on specific milestones within the Ladd School redevelopment project.

Problem 1

In the first textbook exercise, the project has reached month twenty-two. The planned value at this point is 523,000, the earned value is 535,000, and the actual cost incurred is 540,000.

Given:

Actual Cost = 540,000

Planned Value = 523,000

Earned Value = 535,000

Cost Variance: EV minus AC = negative 5,000

Schedule Variance: EV minus PV = positive 12,000

These results show that the project is ahead of schedule because the earned value exceeds the planned value. However, it is slightly over budget because the actual cost is higher than the earned value. This demonstrates how EVA can reveal mixed performance trends that may not be visible through schedule tracking alone.

Problem 2

In the second exercise, a sales project at month five has a planned value of 42,000, an earned value of 39,000, and an actual cost of 34,000.

Given:

Actual Cost = 34,000

Planned Value = 42,000

Earned Value = 39,000

Cost Variance: EV minus AC = positive 5,000

Schedule Variance: EV minus PV = negative 3,000

Cost Performance Index: EV divided by AC is approximately 1.147

Schedule Performance Index: EV divided by PV is approximately 0.929

These results show that the project is under budget because the earned value exceeds the actual cost. However, it is slightly behind schedule because the earned value is lower than the planned value. This example illustrates how EVA provides a balanced view of both cost and schedule performance.

Earned Value Analysis Applied to the Ladd School Project

To illustrate how EVA supports performance monitoring, three milestones from the Ladd School redevelopment were evaluated using the three percent completion methods. These examples demonstrate how different methods influence the interpretation of progress and cost efficiency.

0/100 Method: Certificate of Occupancy

Budget for certificate of occupancy: 200,000

Planned Value at reporting point: 200,000

Earned Value using 0/100: 0

Actual Cost: 150,000

Cost Variance: negative 150,000

Schedule Variance: negative 200,000

Cost Performance Index: 0

Schedule Performance Index: 0

This result reflects the strict nature of the 0/100 method. Although work is underway, no value is recognized until the certificate is fully obtained. This prevents overestimating progress on critical regulatory milestones.

50/50 Method: HUD CDBG Grant Application

Budget for grant application: 100,000

Planned Value at midpoint: 50,000

Earned Value using 50/50: 50,000

Actual Cost: 60,000

Cost Variance: negative 10,000

Schedule Variance: zero

Cost Performance Index: approximately 0.833

Schedule Performance Index: 1.0

The task is on schedule but over budget. The 50/50 method smooths progress recognition while still revealing cost inefficiency. This is useful for administrative tasks that do not lend themselves to granular measurement.

Percent Complete Method: Masonry and Façade Repair

Budget for masonry repair: 1,000,000

Planned Value at reporting point: 600,000

Earned Value at 50 percent complete: 500,000

Actual Cost: 550,000

Cost Variance: negative 50,000

Schedule Variance: negative 100,000

Cost Performance Index: approximately 0.909

Schedule Performance Index: approximately 0.833

The masonry work is behind schedule and over budget. The percent complete method provides a realistic view of progress for long duration construction tasks and highlights areas where corrective action may be needed (Drexel Tech, 2026; LocalCode, n.d.).

Project Management Software Output

To support the earned value calculations presented in this paper, Figure 1 provides a visual representation of task progress from Microsoft Project, including percent complete values for key activities within the Ladd School redevelopment project. Although the earned value calculations in this paper represent specific reporting points for individual milestones, the visual below provides a snapshot of percent complete values within the overall project schedule. Because this project is fictional, the percent complete values shown are illustrative rather than reflective of real progress.

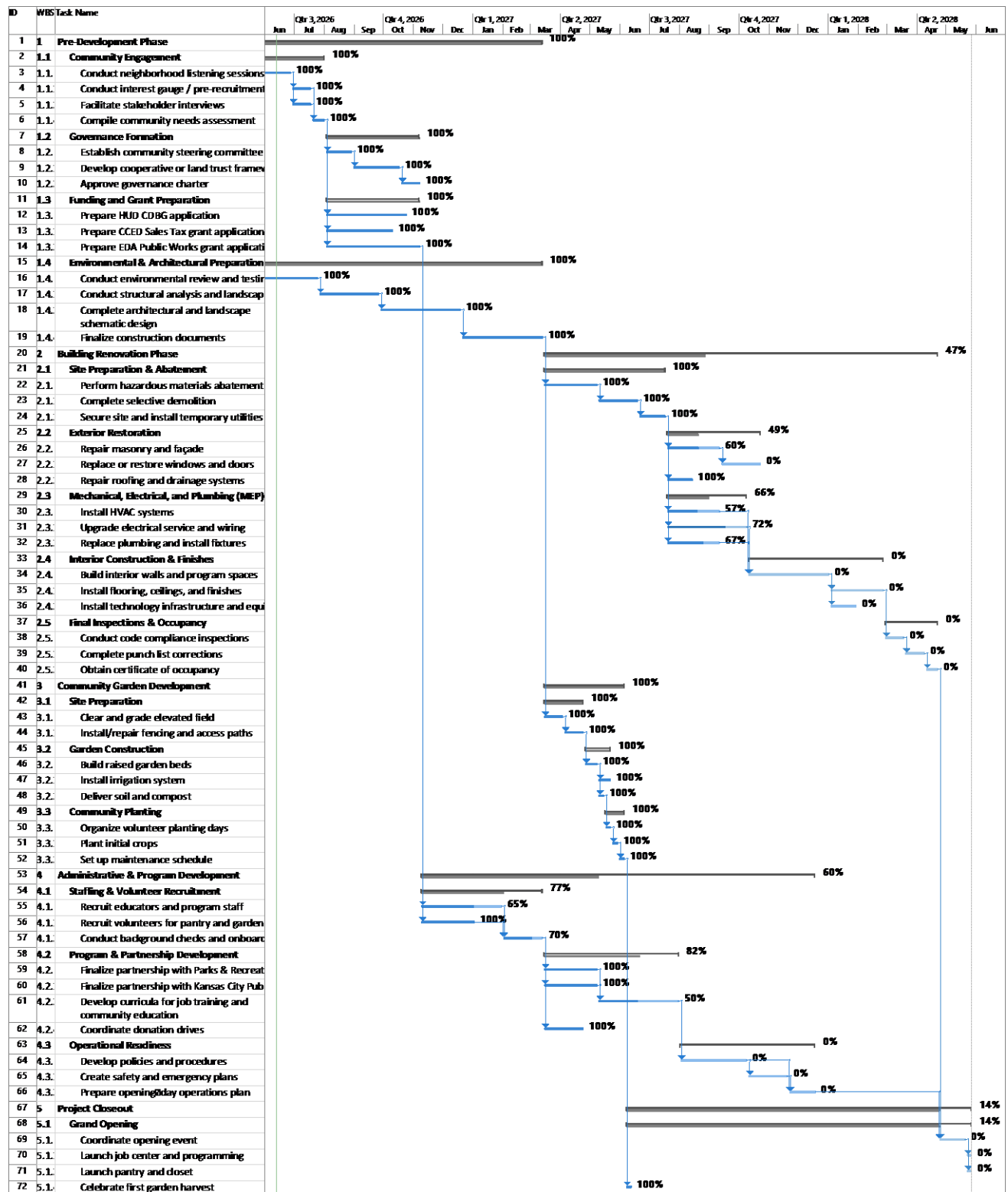


Figure 1.

Microsoft Project percent complete view for the Ladd School redevelopment.

Conclusion

Earned Value Analysis provides a structured and quantitative approach to evaluating project performance. By applying EVA to the Ladd School redevelopment, it becomes clear how different percent completion methods influence the interpretation of progress and cost efficiency. The 0/100 method is appropriate for binary deliverables, the 50/50 method supports documentation-based tasks, and the percent complete method provides detailed insight into construction progress. Together, these methods support accurate forecasting and informed decision making throughout the redevelopment process.

References

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